WaveVis magnetic valve sandbox for EPM geometry and ball pull

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This paper explains the Magnetic Valve tab in WaveVis. The current goal is not pneumatic flow. The current goal is to compare electropermanent-magnet-style geometries and see which layout pulls a steel ball sideways hardest. The model is a fast client-side design sandbox for Alnico, NdFeB, iron pole pieces, coil programming state, flux-line visualization, exterior-air flux-density display, and ball force direction. It is not COMSOL, not FEMM, not measured magnetic force, and not validated material saturation. The paper exists because a physics-looking simulator cannot be allowed to feel like magic.

1 Introduction

An electropermanent magnet combines a high-coercivity permanent magnet, usually NdFeB, with a lower-coercivity programmable magnet, often Alnico. A pulse changes the Alnico magnetization state while the NdFeB stays essentially fixed. When the two magnets align, the external field can be strong. When they oppose, more flux closes internally through soft iron and less leaks outside. This on/off behavior is the reason electropermanent magnets are useful in low-power holding, connectors, actuators, and valves (*1–3*).

The Magnetic Valve tab was originally pulled in too many directions: pneumatic flow, barbed fittings, coil wrapping, moving balls, seats, and COMSOL-like field pictures. The useful design question became simpler: for a given arrangement of NdFeB, Alnico, iron pieces, and a steel ball, which geometry creates the strongest lateral pull on the ball? The current sandbox is therefore a geometry comparison tool. It should help choose relative positions, sizes, grades, pulse settings, and iron-pole placement before a narrower geometry is sent to FEMM, COMSOL, or a physical test.

The major lesson from the implementation cycle is that magnetic field lines are not decorative lines. A field line must be integrated from one field. A heatmap must display one scalar derived from that same field. Iron and the ball must affect the field if the UI implies they do. The density colorbar must be normalized for the exterior-air region when the design question is outside the solid parts. A line family drawn by hand or by a separate helper is not evidence.

2 Implementation Source

Source file	Role
`src/magneticValveModel.ts`	Defines the magnetic components, grade tables, pulse model, EPM circuit score, field grid, flux-line tracing, exterior density display, and ball force estimate.
`src/MagneticValveTab.tsx`	Renders the shared-width dark left inspector, one object picker, selectable/draggable components, direct transform handles, add-iron square/circle actions, compact Alnico state controls, pulse buttons, run and full-screen buttons, canvas, compact black colorbar, force arrow, plain result readouts, and collapsed pulse constants.
`scripts/check-magnetic-valve.cjs`	Unit sanity checks for poles, Alnico states, pulse direction, force direction, standalone fields, iron influence, ball influence, and geometric response.
`scripts/check-magnetic-ui-cdp.mjs`	Browser-level checks for dragging, dynamic field updates, visible flux lines, flux-density rendering, and compact UI behavior.

Table 1. Magnetic Valve implementation map. The paper describes the source model and its limits; the source remains the executable truth.

3 Magnetic Components

The default component set is deliberately small:

- one NdFeB magnet with fixed magnetization;
- one Alnico magnet with programmable state: off, NS, or SN;
- two default soft-iron pole pieces;
- optional added iron square and iron circle primitives;
- one steel ball.

The default values in source are millimeter-scale, with a 34 mm workspace, a 1.55 mm steel ball, a 14 mm NdFeB bar, a 14 mm Alnico bar, and two iron pieces. The material tables are approximate grade controls:

Material setting	Source value used	Meaning
Alnico 5	$B_r = 1.25$ T, switch field 240 kA/m	Low-coercivity programmable magnet approximation.
Alnico 8	$B_r = 0.83$ T, switch field 560 kA/m	Higher coercivity Alnico approximation.
Alnico 9	$B_r = 1.12$ T, switch field 430 kA/m	Intermediate programmed magnet option.
NdFeB N35	$B_r = 1.17$ T	Fixed high-coercivity magnet approximation.
NdFeB N42	$B_r = 1.28$ T	Default fixed magnet approximation.
NdFeB N52	$B_r = 1.43$ T	Higher remanence option.

Table 2. The values are design parameters taken from vendor-grade ranges and simplified for the sandbox (*4*–*6*). They are not measured values for Alan's exact magnets.

4 EPM State Model

4.1 Magnetization states

For a rectangular magnet component j, the sandbox assigns a signed remanence

$$B_{r,j} = p_j B_{r,\text{grade}} s_j,$$

where $p_j \in \{-1, 1\}$ is polarity and $s_j \in [0, 1]$ is programmed strength. For NdFeB, $s_j = 1$. For Alnico:

$$s_j = \begin{cases} 0, & \text{off}, \\ \text{programmedStrength}, & \text{NS or SN}. \end{cases}$$

The source uses polarity to place poles at the ends of the bar. If the magnet is horizontal, NS means one end is north and the other is south according to the signed polarity convention in the source.

4.2 Pulse programming

The pulse model is the simplest useful EPM approximation. A coil pulse creates an approximate magnetizing field

$$H_{\text{pulse}} \approx \frac{NI}{\ell},$$

where N is turns, I is current, and ℓ is the coil length. The source expresses this in kA/m. The programmed fraction is

$$s_{\text{Alnico}} = \text{clamp}\left(\frac{H_{\text{pulse}}}{H_{\text{switch}}}\sqrt{\frac{t_{\text{pulse}}}{0.5\text{ ms}}}, 0, 1\right).$$

Pulse polarity and winding direction set the sign. This is not a hysteresis curve. It is a control knob that makes pulse direction, current, turns, duration, and Alnico grade matter in the same direction as the physical intuition.

4.3 Circuit score

The EPM circuit state uses signed magnetic moments

$$M_j = A_j B_{r,j},$$

where A_j is the magnet's 2D area proxy. The net moment is

$$M_{\text{net}} = M_{\text{NdFeB}} + M_{\text{Alnico}}.$$

The app combines this with soft-iron face alignment, closure score, gap distance, and pole orientation to name a qualitative circuit state: aligned, opposed/internal, near cancel, open layout, attracting gap, or repelling gap. This result strip is a guide. It is not a magnetic circuit reluctance solve.

5 Field Model

5.1 Point-source display field

For force and fast line/source behavior, the source keeps a pole-pair field. A north pole contributes a field away from the pole:

$$\mathbf{B}_N(\mathbf{x}) = k\frac{\mathbf{x} - \mathbf{x}_N}{(\|\mathbf{x} - \mathbf{x}_N\|^2 + a^2)^{3/2}},$$

and a south pole contributes a field toward the pole:

$$\mathbf{B}_S(\mathbf{x}) = k\frac{\mathbf{x}_S - \mathbf{x}}{(\|\mathbf{x} - \mathbf{x}_S\|^2 + a^2)^{3/2}}.$$

The softening length a prevents singular pixels. A magnet is the sum of one north and one south pole. This model is useful for local direction and for keeping standalone magnets visible, but it is not a finite-element solution.

5.2 Iron focusing and guidance

Soft iron is represented by two effects. First, nearby iron increases local field magnitude through a bounded focus factor:

$$g(\mathbf{x}) = 1 + \mathrm{clamp}\left(\sum_i \frac{\alpha A_i}{\left\|\mathbf{x} - \mathbf{x}_{i,\mathrm{iron}}\right\|^2 + \beta}, 0, g_{\max}\right).$$

Second, iron adds a guidance vector toward the nearest point on an iron component:

$$\mathbf{B}_{\mathrm{guide}}(\mathbf{x}) = \sum_i \gamma_i(\mathbf{x})\,\|\mathbf{B}_0(\mathbf{x})\|\,\frac{\mathbf{q}_i(\mathbf{x}) - \mathbf{x}}{\|\mathbf{q}_i(\mathbf{x}) - \mathbf{x}\|},$$

where $\mathbf{q}_i$ is the nearest point on the iron. This is a bounded flux-guide approximation. It makes moving iron affect the displayed field and ball force, but it does not model saturation, hysteresis, eddy currents, or anisotropic steel.

5.3 Induced steel ball field

The steel ball is modeled as an induced dipole aligned with the field at the ball center. Let

$$\mathbf{B}_b = \mathbf{B}(\mathbf{x}_b)$$

without the ball's own field. If $\|\mathbf{B}_b\|$ is nonzero, the induced axis is

$$\hat{\mathbf{u}} = \frac{\mathbf{B}_b}{\|\mathbf{B}_b\|}.$$

The ball contributes a small pole pair at

$$\mathbf{x}_{N,b} = \mathbf{x}_b + 0.55r\hat{\mathbf{u}}, \qquad \mathbf{x}_{S,b} = \mathbf{x}_b - 0.55r\hat{\mathbf{u}},$$

with strength proportional to $\|\mathbf{B}_b\|$, ball radius squared, and ferrous-strength slider. This is a design approximation for a high-permeability sphere; it is not a closed-form sphere solution or FEM result (*7*, *8*).

6 Grid Magnetostatic Approximation

For the colored density field and red flux lines, the source builds a 2D finite-difference scalar-potential-style grid. The current grid is 112×96 for the solver and 512×328 for displayed exterior density samples. Each grid cell stores a relative permeability μ_r and remanence vector $\mathbf{B}_r = (B_{rx}, B_{ry})$.

The simplified governing form is

$$\nabla \cdot (\mu_r \nabla \phi) = \nabla \cdot \mathbf{B}_r,$$

then

$$\mathbf{B} = -\mu_r \nabla \phi + \mathbf{B}_r.$$

The source assembles the right-hand side with centered differences of $\mathbf{B}_r$, solves ϕ by successive over-relaxation, and samples $\mathbf{B}$ by finite differences. Interface coefficients use harmonic means:

$$\mu_{ab} = \frac{2\mu_a \mu_b}{\mu_a + \mu_b}.$$

This is only a coarse, interactive field approximation. Proper validation should use FEMM or COMSOL with material curves, boundary conditions, mesh convergence, and three-dimensional geometry (*8*, *9*). The WaveVis grid is still useful because every visual mark comes from one computed field instead of separate hand-drawn line families.

7 Flux Lines

Flux lines are traced by integrating the normalized grid field:

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}s} = \pm\frac{\mathbf{B}(\mathbf{x})}{\|\mathbf{B}(\mathbf{x})\|}.$$

The source traces forward and backward from a seed point, uses a midpoint step, smooths the resulting polyline lightly, and rejects lines that are too short, outside the stage, too jagged, or crossing a detected repelling gap. Lines are seeded around magnet poles, solid faces, the open stage, and the ball. They are deduplicated and limited so the screen stays responsive.

The field lines should follow these sanity cases:

- a standalone magnet shows lines in the open page, even when iron is far away;
- unlike facing poles produce bridging lines through the gap;
- like facing poles produce a low-line-density repelling gap, with lines bending away rather than drawing fake bridges;
- moving iron changes the field lines because iron changes μ_r and guidance;
- moving the ball changes nearby field lines because the ball is an induced high-permeability object;
- line density and seed placement are visualization choices, but line direction comes from the same **B** field.

8 Exterior-Air Flux Density

Alan specifically wanted to judge the field around the components, not inside the magnets or iron. The source therefore treats the heatmap as an exterior-air display. For each display cell, the renderer checks nine sample points against the current solid domains. A sample inside a solid is not evidence. A sample in the numerical material-transition guard at a solid boundary is also not allowed to set the exterior-air scale; it is pushed outward to the display-air clearance before $|B|$ is read. A cell becomes scale-setting exterior evidence only when it is outside solids, outside the interface guard, and not a partial solid overlap. Boundary/interface cells that are visibly outside an object are display-filled from nearby trusted exterior air, but they are flagged as fill, not evidence. Thus red that appears just outside an iron edge must come from trusted exterior-air samples, not from material-domain density under the cell.

The raw displayed scalar for a visible exterior cell is

$$r_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \left\| \mathbf{B}(\mathbf{x}_{ij}) \right\| .$$

The color scale is defined from trusted exterior-air cells only. Let S be the set of raw values from cells that pass the exterior evidence test. Solid interiors, partial solid-overlap cells, interface-guard cells, transparent hardware cells, and hidden component values do not set the colorbar. The low, shoulder, and high exterior reference values are

$$r_{\text{low}} = P_{1.5\%}(S), \quad r_{\text{shoulder}} = P_{72\%}(S), \quad r_{\text{high}} = P_{99.2\%}(S).$$

The compression scale is

$$s = \max\left(0.72(r_{\text{shoulder}} - r_{\text{low}}),\ 0.050 r_{\text{high}},\ \epsilon\right),$$

and each exterior value is compressed as

$$c_i = \log\left(1 + \frac{\max(0, r_i - r_{\text{low}})}{s}\right).$$

The normalized display value uses robust percentiles of the compressed exterior set C:

$$\hat{c}_i = \operatorname{clamp}\left(\frac{c_i - P_{1\%}(C)}{P_{98.2\%}(C) - P_{1\%}(C)}, 0, 1\right).$$

The rendered relative color is lifted slightly above pure black and gamma-adjusted for readability:

$$v_i = 0.14 + (1 - 0.14)\hat{c}_i^{0.68}.$$

This prevents high internal solid values and singular edge samples from making every outside region dark. It also keeps the exterior view useful: air near active pole pieces and the ball gap should reach the red/yellow end of the relative scale, while the field around nearby iron, magnets, and the ball should step through orange, yellow, green, and blue instead of leaving almost all exterior air in the lowest blue band.

The opposite failure is also rejected. The exterior display must not flatten a large continuous air region into a single maximum-red shelf. In the current gate, the browser and model checks count high, hot, and max-red exterior cells. Red is reserved for the strongest exterior tail near active pole faces, iron edges, and the ball gap; nearby strong-but-not-maximum regions must still retain orange, yellow, green, and blue variation.

The prior failure class was a visualization artifact: coarse cells, partial solid coverage, projected samples, suppression factors, mask interpolation, and normalization caused rectangular halos, black rings, or distinct fake bands around iron and the ball. Those halos looked like real flux-density structure but moved in jumps as components moved. The current rule is that exterior density must move continuously with the geometry, must not plot solid-interior values as outside-air evidence, must not make exterior-adjacent cells transparent, must not paint material-domain values underneath opaque hardware, and must not create a separate visual material around solid boundaries. Boundary/interface cells do not set the scale; they may only receive a display color filled from trusted exterior air. Fully solid interiors are transparent in the density raster. The SVG no longer uses a separate hardware mask or ball-clear path, because that layer produced black component-shaped holes. The browser gate verifies `interfaceEvidence` = 0, nonzero trusted boundary fill, no density mask, and zero density alpha at the centers of NdFeB, Alnico, both iron pieces, and the ball. Air outside the hardware is air; material-domain density is never promoted into that air just because a raster cell touches a part.

9 Ball Force Estimate

The force on a small ferromagnetic object in a field gradient is approximated as proportional to the gradient of field energy. WaveVis uses

$$U(\mathbf{x}) = \|\mathbf{B}(\mathbf{x})\|^2 ,$$

and a centered finite difference at the ball:

$$\frac{\partial U}{\partial x} \approx \frac{U(x+h,y) - U(x-h,y)}{2h}, \qquad \frac{\partial U}{\partial y} \approx \frac{U(x,y+h) - U(x,y-h)}{2h}.$$

The displayed force vector is

$$\mathbf{F}_{\text{display}} = k_f V_b s_b \begin{bmatrix} \partial U/\partial x \\ \partial U/\partial y \end{bmatrix},$$

where V_b is represented by a diameter-cubed scale and s_b is the ferrous-strength slider. The force is clamped to keep the UI stable. The units are presented as design-scale newtons, not measured force. A real force claim requires a force gauge, calibrated material, and measured geometry.

10 Known Limits

The Magnetic Valve tab intentionally does not claim:

- exact 3D magnetostatics;
- material saturation;
- Alnico hysteresis loop accuracy;
- measured B in tesla;
- measured force in newtons;
- coil heating;
- eddy currents;
- pneumatic flow;
- seat leakage;
- validated valve closure.

It does claim a narrower thing: changing magnet/iron/ball geometry changes one approximate field, and the approximate field is used consistently for field lines, exterior density, and ball-pull direction.

11 Verification Cases

The correct test set is larger than a screenshot. The simulator must pass:

1. **Standalone field:** NdFeB alone shows visible field lines.
2. **Alnico off:** Alnico contributes no remanent field.
3. **Alnico NS/SN:** switching Alnico reverses its poles and changes external field direction.
4. **Pulse sign:** pulse + and pulse - program opposite Alnico directions when winding is fixed.
5. **Winding reversal:** reversing winding flips the programming response.
6. **Like-pole gap:** like poles repel; the gap should not be drawn as an attractive bridge.
7. **Unlike-pole gap:** unlike poles produce bridge-like field lines.
8. **Iron movement:** moving iron changes field lines, density, and ball force.
9. **Ball movement:** moving the ball changes local field and force direction.
10. **Exterior density:** the colorbar is normalized from outside-air values, not solid interiors.
11. **Solid-domain separation:** the density layer is transparent under NdFeB, Alnico, iron, and the visible ball body, so only exterior air sets the view.
12. **Exterior scale:** pole-adjacent exterior air must render high on the relative colorbar, exterior air must not render black, and mid/far exterior air must fall off rather than leaving the page mostly blue/black.
13. **Exterior saturation:** max-red exterior area must stay bounded so strong regions do not become a flat red shelf.
14. **Mask artifact:** no rectangular, circular, transparent black, projected-air, or artificial boundary-band halo from density cells is allowed to be mistaken for physics.
15. **Live drag:** lines and density update during drag, not only after pointer release.
16. **Performance:** drag remains responsive with reduced line count and bounded density resolution.
17. **Direct geometry editing:** added iron square/circle primitives and selected-object transform handles work without forcing side-panel-only resizing.

12 Discussion

The magnetic sandbox should be used to compare geometries, not to prove final performance. A good result is a layout where the force arrow consistently points toward the desired ball motion, the side-pull metric improves relative to alternatives, the flux lines are physically plausible under the listed sanity cases, and the exterior density map shows concentration near the ball gap without display artifacts. A bad result is one where the picture looks dramatic but the field lines disappear for standalone magnets, fake multiple line families appear, the colorbar is dominated by solid interiors, or dragging components produces jumps.

The next validation path is not more visual tuning. It is to choose a small set of candidate geometries and run a real magnetostatic solve in FEMM or COMSOL, then measure ball pull experimentally. WaveVis should narrow the search, not replace validation.

13 Data availability

No measured magnetic data are included. The model values are source constants and vendor-grade approximations. Any future measured pull-force data should be added as a separate dataset with magnet grade, dimensions, steel type, ball diameter, gap, and measurement method.

14 Code availability

The active model is `src/magneticValveModel.ts`. The active UI is `src/MagneticValveTab.tsx`. The public app serves the compiled bundle and this PDF through `public/proofs/` after deployment.

15 Additional information

This paper supersedes the earlier pneumatic-valve framing for the current V1. Pneumatics, tube seats, barbed fittings, and flow cutoff are not active requirements in the current Magnetic Valve tab. They are historical context only unless reopened.

References

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